the decentralized consolidation model would allow more timely delivery of consolidated market data.[662] Another commenter said the proposal's content and latency reforms "go a long way."[663]

One commenter stated that the proposal's changes to latency would "modernize market data infrastructure."[664] This commenter said that competition among competing consolidators would reduce geographic and aggregation latency.[665] Other commenters also noted the proposed decentralized consolidation model's potential beneficial effects on geographic latency.[666] One commenter stated that the proposal would reduce geographic latency because exchange data would no longer be aggregated by the exclusive SIPs in two locations, and competing consolidator subscribers could receive consolidated data within the data center of the competing consolidator.[667] Two commenters said that the proposed decentralized consolidation model could reduce the geographic, aggregation, and transmission latency differentials between proprietary market data feeds and SIP data.[668] Other commenters also noted the proposed

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662 See DOJ Letter at 4.

663 AHSAT Letter at 1.

664 SIFMA Letter at 1.

665 See id. at 11. See also NovaSparks Letter at 1 (stating that competition will encourage competing consolidators to deliver excellent performance).

666 See MEMX Letter at 6, 7, 8; ICI Letter at 10; BlackRock Letter at 5.

667 See ICI Letter at 10.

668 See MEMX Letter at 6, 8; BlackRock Letter at 5. See also NBIM Letter at 6 (stating that it uses direct feeds to reflect the "physical reality" of the broker-dealers whose performance it needs to evaluate and that the proposal would provide an opportunity for competitive processors located in the same data centers as most institutional broker-dealers to emerge).

decentralized consolidation model's potential to reduce both the latency and the content differentials between proprietary market data feeds and SIP data.[669]

However, several commenters questioned whether the decentralized consolidation model could meaningfully impact the latency of consolidated market data.[670] One commenter said that there was no guarantee that competition would result in improved latency.[671] Another said that exchanges could increase the latency gap between proprietary data and consolidated market data with frequent upgrades.[672] One commenter stated that the proposal failed to explain how the decentralized consolidation model would reduce latency and asked the Commission to explain why the proposed model is preferable to the distributed SIP alternative, which would address geographic latency.[673] Another commenter said that the proposed decentralized consolidation model is inconsistent with the Commission's obligations under the APA because it is not based on current market conditions and relies instead on "outdated discussions and panelist comments"

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669 See ACS Execution Services Letter at 5 (stating that the model would reduce content and latency differentials between SIP and proprietary market data); DOJ at 2–3, 4 (supporting the proposal's efforts to address the granularity and latency differentials between SIP and proprietary market data).

670 See Cboe Letter at 23; Citadel Letter at 5; STANY Letter II at 5, 6; NYSE Letter II at 11, 22, 23; Nasdaq Letter IV at 49; Angel Letter at 18, 19; TD Ameritrade Letter at 12; FINRA Letter at 8; IDS Letter I at 15; Data Boiler Letter II at 2; Proof Trading Letter at 1.

671 See TD Ameritrade Letter at 12.

672 See Data Boiler Letter II at 2.

673 See Nasdaq Letter IV at 49.

from the Market Data Roundtable.[674] The commenter said that changes to market data infrastructure and governance have since reduced the latency differentials.[675]

The Commission believes that fostering a competitive environment for the collection, consolidation, and dissemination of consolidated market data will result in such data being delivered to market participants in a decentralized manner with improved geographic, aggregation, and transmission latencies. With respect to geographic latency, unlike the current exclusive centralized consolidation model, the decentralized consolidation model will allow the direct delivery of each SRO's market data to competing consolidators and self-aggregators, and competing consolidators may be located in the same data center as their subscribers. This stands in stark contrast to today's model where (a) one consolidator is located in one centralized data center while (b) a significant number (in some cases a majority) of subscribers are located in different data centers, and (c) each SRO's market data is required to travel to the one centralized location to be aggregated, prior to (d) traveling to yet another data center for receipt and use by subscribers.[676] In the decentralized consolidation model, SRO data will no longer be required to

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674 NYSE Letter II at 9, 10, 23. Similarly, one commenter stated that the Commission solely relied on comments from the Market Data Roundtable to support its belief that the decentralized consolidation model would reduce transmission latency differentials between SIP and proprietary market data. See Nasdaq Letter IV at 45.

675 The commenter said that the Commission has ignored "the impact of significant changes to the SIP infrastructure already implemented by the SROs and to the governance of the national market systems that the Commission recently imposed, while overlooking the impressive performance of the existing system in a time of extreme market volatility." NYSE Letter II at 10.

676 Today, each exclusive SIP must collect data from geographically dispersed SRO data centers, consolidate the data, and then disseminate the consolidated data from the exclusive SIP's location to end-users, which are often in other locations, in a hub-and-

travel to a separate central location for consolidation by an exclusive SIP. Consolidation could occur at the data center where a data end-user is located instead of occurring only at the CTA/CQ SIP and the Nasdaq UTP SIP data centers. As one commenter stated, the physical location of a processor is critical.[677]

Furthermore, competition will incentivize competing consolidators to minimize latency and improve aggregation and transmission performance and services for consolidated market data products through the use of low-latency aggregation and transmission technologies.[678] Competing consolidators and self-aggregators will not be restricted to the transmission methods mandated by the Equity Data Plans, and competing consolidators will compete with each other based on the efficiency of their aggregation of raw SRO data to generate consolidated market data. In contrast to today's non-competitive exclusive SIPs, the Commission believes that competing consolidators will be incentivized to make continued improvements.[679] For example,

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spoke form of centralized consolidation that creates additional latency. See Proposing Release, 85 FR at 16765.

677 See NBIM Letter at 4.

678 See infra Section V.C.2(c) (discussing the effect of the decentralized consolidation model on innovation in data delivery and reducing latency differentials). Although the exclusive SIPs have reduced their aggregation latencies and made other improvements, as the Commission stated above, there is currently no competition for consolidated market data, and the technology for the distribution of SIP data has continued to meaningfully lag behind technologies utilized across the private competitive data landscape. See supra Section III.B.2.

679 See text accompanying notes 602–603. The Commission notes that the Nasdaq UTP SIP revised its technology in the fourth quarter of 2016 to lower quote latency at the 99th percentile from 5,393 microseconds to 28 microseconds. In the same quarter, the CQS SIP's 99th percentile of latency was 1,570 microseconds and it did not reduce that latency to below 100 microseconds until the third quarter of 2020. See Nasdaq UTP Q3 2020 – September Tape C Quote Metrics, available at http://utpplan.com/DOC/UTP_Website_Statistics_Q3-2020-September.pdf (last accessed

competing consolidators will be incentivized to minimize the amount of time it takes to aggregate consolidated market data products;[680] reduce their transmission latency[681] (e.g., by offering wireless connectivity through microwave or laser technology, currently offered by exchanges[682]); reduce connectivity latency (e.g., by offering field-programmable gate array ("FPGA") services[683]); lower connectivity fees; enhance customer service; and enhance their technology and services to remain competitive. Competing consolidators providing consolidated market data products to clients for electronic trading will likely compete along all of these lines, similar to the manner in which the providers of proprietary data products have competed. Further, self-aggregators will be able to better utilize technologies to perform their aggregation and transmission functions.

Nov. 27, 2020); Key Operating Metrics of Tape A&B U.S. Equities Securities Information Processor (CTA SIP), available at https://www.ctaplan.com/publicdocs/ctaplan/CTAPLAN_Processor_Metrics_3Q2020.pdf (last accessed Nov. 27, 2020).

680 See SIFMA Letter at 11; BlackRock Letter at 5.

681 See ICI Letter at 10; BlackRock Letter at 5.

682 See, e.g., ICE Global Network: New Jersey Metro, available at https://www.theice.com/market-data/connectivity-and-feeds/wireless/new-jersey-metro (last accessed Nov. 27. 2020); Nasdaq, Wireless Connectivity – Metro Millimeter Wave Frequently Asked Questions, available at https://www.nasdaq.com/docs/2020/01/15/Metro_Millimeter_Wave_FAQ.pdf (last accessed Nov. 27, 2020).

683 See NovaSparks Letter at 1. See also Nasdaq Equity Trader Alert #2015-194, "Nasdaq Reintroducing FPGA Order Entry Ports, Announcing Port and Pricing Updates for 2016," available at https://www.nasdaqtrader.com/TraderNews.aspx?id=ETA2015-194 (last accessed Nov. 27, 2020); Flanagan, Terry, Co-Location: How Close Can You Get?, MarketsMedia (Dec. 27, 2012), available at https://www.marketsmedia.com/co-location-how-close-can-you-get/ (explaining the use of FPGA to reduce co-location latencies).

One commenter asserted that the proposed decentralized consolidation model assumed that competing consolidators would specialize in lower latency data but said that this assumption was accurate only if the SROs from which they receive data can offer low-latency connectivity.[684] The commenter noted that the proposal could result in the discontinuation of low-latency connectivity options by SROs and said that the Commission did not assess the impact of the proposal on this connectivity market.[685] The Commission believes that this comment fails to recognize the ways in which the proposal addressed the connectivity market. In specifying "by the same means, and on the same terms," at a minimum, the Commission has prohibited an SRO from providing superior connectivity for proprietary data products than it provides for NMS data. The Commission further addresses these concerns below.[686]

Some commenters stated that competing consolidators would not eliminate geographic latency.[687] One commenter stated that geographic latency would still exist despite implementation of the proposed decentralized consolidation model.[688] The commenter stated that incremental reductions to transmission latency would be the most the decentralized consolidation model could achieve but that the Commission failed to analyze whether such reductions would be worth the cost of the proposal.[689] Another commenter stated that competing

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684 See IDS Letter I at 15.

685 See id.

686 See infra Section III.B.9.

687 See Citadel Letter at 5; STANY Letter II at 6; NYSE Letter II at 11, 23.

688 See NYSE Letter II at 23.

689 See id. at 11.

consolidators would still be subject to geographic and operational latency, which would result in competing consolidators located within the same data center disseminating differing prices.[690]

Other commenters stated that the proposed decentralized consolidation model would reduce the latencies associated with the dissemination of SIP data.[691] The Commission agrees and believes that the model will significantly reduce geographic latency because, as described above, it will allow consolidation of market data to occur at the data center where a data end-user is located and end the consolidation of data at a single location where end-users may not be located. While full elimination of geographic latency for NMS data is impossible in a marketplace where different markets are located in different geographic locations, the reduction of latency caused by a centralized consolidation requirement is desirable and will result in significant latency benefits. If a competing consolidator chooses not to provide a consolidation service in all of the data centers of its users, the Commission believes the users will still benefit from reduced aggregation and transmission latencies resulting from competition among competing consolidators.

Finally, one commenter said that the current latency of the exclusive SIPs is sufficient for agency trading and doubted that any latency improvements would benefit long-term investors.[692] However, several commenters representing long-term investors expressed the view that the current latency of the SIPs was not sufficient to meet their needs.[693] While the current latency of the exclusive SIPs may be sufficient for some retail investors and other visual consumers of

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690 See Angel Letter at 18.

691 See AHSAT Letter at 1; BlackRock Letter at 5; DOJ Letter at 2–3, 4; Fidelity Letter at 10; MEMX Letter at 3, 6, 7, 8; SIFMA Letter at 1, 11; Wellington Letter at 1; ICI Letter at 10; ACS Execution Services Letter at 5; Better Markets Letter at 3.

692 See Proof Trading Letter at 1.

693 See, e.g., Capital Group Letter at 2, 4; Fidelity Letter at 2; State Street Letter at 2.

market data, the Commission believes that the reduction in latency should enhance trading by the brokers who service retail investors by allowing them to evaluate the markets quickly, adjust their quotes, trade more efficiently and competitively, and facilitate best execution. Furthermore, the addition of new data content in consolidated market data and the competitive environment fostered by the decentralized consolidation model may allow agency brokers to purchase consolidated market data products, which may be offered at a lower cost than current proprietary data, rather than proprietary data feeds, which could result in cost savings for investors.

While some commenters stated the proposed decentralized consolidation model would have little, if any, impact on latency,[694] other commenters said that the decentralized consolidation model would perpetuate latency differentials.[695] One commenter stated that competing consolidators would initiate a "costly arms race in speed,"[696] resulting in major market participants complaining about having to pay a premium for the fastest consolidator.[697] One commenter said nothing in the proposal would address the Commission's concerns expressed in the Proposing Release[698] about a "two-tiered market data environment"[699] and

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694 See supra note 670.

695 See Nasdaq Letter IV at 8, 23–24; NYSE Letter II at 22, 23; STANY Letter II at 6; Angel Letter at 19; FINRA Letter at 8–9.

696 Angel Letter at 19.

697 See id.

698 See Proposing Release, 85 FR at 16767–8.

699 Nasdaq Letter IV at 23–24. This commenter also said that the "two-tiered" environment adds no cost to the majority of traders and believed that SIP data is sufficient for human traders who would not benefit from expensive infrastructure and that professional traders tend to opt for custom solutions rather than buying the same products anyway. Nasdaq Letter III at 5.

argued that competing consolidators would create a multi-tiered market where market participants would be charged more for better products and faster services.[700] Further, this commenter stated that competing consolidator subscribers, such as retail investors, would be at a latency disadvantage to self-aggregators that can generate an NBBO faster.[701] This commenter also said that competing consolidators could even start a "new fragmentation war" for latency-sensitive subscribers that need to be co-located near their competing consolidator.[702]

The Commission believes that competing consolidators, based on subscriber demand, will develop different consolidated market data products for their subscribers and will compete on the basis of latency, resiliency, products and services offered, and other factors, including price. Subscribers also will be able to evaluate competing consolidators on the basis of system availability, network delay statistics, and data quality and system issues that will be publicly available in the Form CC and the performance statistics and operational information required to be disclosed by competing consolidators on a monthly basis by Rule 614. Different subscribers and trading applications may prioritize these factors differently. The Commission recognizes that there will be different needs for different participants and applications. However, the Commission does not believe that the realm of such differentiation and innovation should be exclusively limited to proprietary data products. As noted above, some commenters believed that the decentralized consolidation model would perpetuate latency differentials.[703] Although there may be differences in the latencies among competing consolidators, the Commission

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700 See Nasdaq Letter IV at 8.

701 See id. at 8, 42.

702 Id. at 26.

703 See Nasdaq Letter IV at 8, 23–24; NYSE Letter II at 22, 23; STANY Letter II at 6; Angel Letter at 19; FINRA Letter at 8–9.

believes the decentralized consolidation model will result in a net benefit in overall improved latencies for users of consolidated market data relative to the current model, and competitive market forces should reduce the likelihood of an unlevel playing field.

Two commenters stated that self-aggregators would have a latency advantage over competing consolidators, which would continue a two-tiered market data environment despite the presence of competitive forces.[704] One of the commenters said that self-aggregators would continue to obtain and use market data faster than subscribers of competing consolidators.[705] The commenter also said that the proposal lacked an analysis of the latency advantages of self-aggregators over competing consolidators.[706] Another commenter said the decentralized consolidation model would "institutionalize latency inequities" through the use of self-aggregators and competing consolidators.[707] The commenter stated that the latency advantage of self-aggregators over competing consolidators was not actually minor, nor did it believe that competing consolidators could minimize the latency differences.[708] This commenter suggested that the Commission should either require the SROs to delay provision of market data to self-aggregators or allow only competing consolidators to provide consolidated market data.[709] One

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704 See NYSE Letter II at 22, 23; STANY Letter II at 6.

705 See NYSE Letter II at 23. See also Healthy Markets Letter I at 2–3.

706 See NYSE Letter II at 23.

707 FINRA Letter at 8.

708 See id. at 8. The commenter also said that competing consolidators that aggregate data for themselves would have a latency advantage over their subscribers. Id.

709 See id. at 8–9. See also Healthy Markets Letter I at 3 (recommending that an exchange that wished to send data to its customer be required to do so "through an affiliate that would receive the same data, at the same time, on the same terms, and at the same cost as any competing SIP distributor").